Weak-Signal Interpretation for Low-Altitude Drone Detection

Toward a Stratified, Deviation-Governed Sensing Architecture

Stephen A. Putman
putmanmodel@pm.me

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Abstract

Low-altitude small-drone detection is difficult not only because targets are physically small, but because sensing occurs under noisy, cluttered, and often ambiguous conditions in which weak cues must be combined without collapsing prematurely into high-confidence claims. This paper proposes a stratified detection architecture that treats drone sensing as a weak-signal interpretation problem rather than as a single-stage classification task. The architecture separates bounded observation, short-horizon salience, structural deviation evaluation, and governed consequence. Weak acoustic, visual, thermal, RF, and motion-derived partials are represented provisionally; compared against short-horizon and scope-specific baselines; escalated under persistence, convergence, or meaningful deviation; and prevented from directly becoming durable state or high-impact action without additional evidence. The proposal is architectural, defensive, and simulation-oriented. It does not present interception tactics or offensive optimization. Its contribution is a disciplined way to handle low-confidence environmental cues under uncertainty while reducing both false indifference and false certainty.

1. Introduction

Low-altitude small-drone detection is often framed as a sensor problem: better radar, better optics, better classifiers. That framing is incomplete. In cluttered environments, many early indicators are weak, partial, and ambiguous. A system may receive an acoustic irregularity, a low-confidence motion pattern, a transient thermal inconsistency, or a bearing shift that differs from local ambient conditions without yet justifying a durable object claim. The architectural question is therefore not only whether a drone can be detected, but how weak signals should be represented, escalated, and governed before they harden into persistence or consequence.

This matters because the earliest useful cues are often not cleanly resolved objects. They are departures from a local corridor of expectation. A sound may differ from wind in periodicity or modulation. A motion trace may persist slightly longer than ambient clutter. A thermal speck may appear coherent enough to merit attention while remaining too weak to deserve a confident label. Under those conditions, forcing the system into immediate binary classification is often the wrong move. The more disciplined alternative is to allow weak cues to matter provisionally without permitting them to collapse directly into durable state or stronger consequence.

The present paper proposes a narrow architectural response. It treats low-altitude drone sensing as a weak-signal interpretation problem under explicit stratification. Bounded findings are emitted first. Candidate local meanings are formed second. Attentional escalation occurs only when deviation becomes informative. Governed promotion follows only under stronger evidence, persistence, or convergent support across dimensions. The goal is not omniscience, total certainty, or a complete operational doctrine. The goal is a sensing architecture that handles uncertainty coherently.

This proposal is defensive and simulation-oriented. It is not a tactics paper, an interception paper, or a deployment claim. Its contribution is architectural discipline under clutter: a way to preserve sensitivity to weak signals while reducing both false indifference and false certainty.

2. Problem Framing: Why Small-Drone Detection Is an Interpretation Problem

Low-altitude small drones are difficult not only because they are small, but because they are often encountered through imperfect, overlapping, and environmentally noisy cues. In practice, the first available evidence may be acoustic, visual, thermal, motion-derived, RF-related, or some weak combination of these. Any one cue may be insufficient on its own. Worse, many environments already contain structured noise: wind, road traffic, rotating machinery, birds, thermal shimmer, intermittent reflections, or sensor dropout. The system is therefore not merely asked to decide whether a drone is present. It is asked to decide what to do with an ambiguous pattern that differs from ambient conditions in a structured but still underdetermined way.

That makes early sensing an interpretation problem. The system must compare local findings against a bounded corridor of expectation and decide whether deviation is trivial, informative, or serious enough to justify broader inspection. A low-flying drone may first appear not as a resolved object, but as a bounded anomaly relative to a local baseline. If the system ignores such weak departures, it loses useful early structure. If it overcommits too early, it turns provisional signal into brittle certainty. Both errors are costly.

A more disciplined architecture therefore needs somewhere for uncertainty to live. It must be able to represent a weak cue, treat it as meaningful enough to influence near-term attention, and still refuse to treat it as a durable track or stronger response trigger until additional conditions are met. In this sense, the relevant design question is not simply “can this sensor detect a drone?” but “how should the system behave when evidence is incomplete, weakly convergent, or temporarily ambiguous?”

This paper answers that question by treating low-altitude drone sensing as a stratified weak-signal pipeline rather than as an immediate object declaration. The premise is modest: in cluttered sensing conditions, good early behavior depends less on perfect first-pass certainty than on disciplined handling of partial evidence.

3. Core Claim and Design Principle

The central claim of this paper is simple: low-altitude drone detection should be treated as a weak-signal interpretation problem, not only as a single-stage classification problem.

This claim does not reject sensors, classifiers, or conventional fusion methods. It reframes their early-stage use. In clutter-heavy environments, many cues are too partial to deserve immediate commitment. They are still useful, but their usefulness lies first in how they alter local attention, candidate interpretation, and evidence-seeking behavior. The architecture should therefore preserve a strict separation between four stages: representation, local interpretation, attentional escalation, and governed consequence.

Representation comes first. A cue cluster is encoded in bounded form rather than forced immediately into a resolved identity claim. Local interpretation comes second. The represented pattern is compared against current expectations for that scene, location, channel, or horizon. Attentional escalation comes third. If deviation is negligible, the system remains cheap and local. If deviation becomes informative, scope widens slightly through reversible means such as reweighting, additional checking, or cross-sensor cueing. Governed consequence comes last. Even a strong local pattern does not automatically deserve durable promotion, system-wide persistence, or stronger action.

The governing design principle is simple: weak signals should be allowed to influence attention before they are allowed to justify stronger commitment. In low-altitude drone sensing, this means an acoustic irregularity, bearing shift, motion persistence, or weak multi-sensor blend can raise attention and alter the next inference step without immediately becoming a durable track, a system belief, or a high-impact trigger.

This is a defensive sensing proposal, not a tactical or offensive one. It does not describe interception methods, swarm defeat logic, or target engagement policy. Its contribution is narrower and more architectural: a disciplined way to handle uncertain environmental cues so that provisional significance remains possible without premature certainty.

4. Stratified Detection Architecture

The proposed architecture treats low-altitude drone sensing as a layered interpretation pipeline rather than as a single-stage recognition event. This is not an argument against sensors or classifiers. It is an argument for preserving architectural boundaries around what each stage is allowed to do. In cluttered environments, early cues often arrive in weak, partial, or cross-modal form. If every stage is allowed to behave as though its local signal were already a resolved object claim, the system becomes brittle. A more disciplined architecture separates bounded observation, provisional interpretation, bounded response shaping, and governed promotion.

At the first layer, the system receives bounded findings from individual sensing paths. These findings are deliberately narrow. An acoustic channel might emit a periodicity anomaly, an unusual modulation pattern, or a directional consistency estimate. A visual pathway might emit low-confidence motion persistence or small-object track continuity. A thermal pathway might emit a localized moving inconsistency. An RF-related subsystem might emit presence, absence, or irregularity information where relevant. None of these outputs should be treated as a durable object claim on their own. Their role is to preserve local structure in a form that downstream layers can compare, combine, and challenge.

At the second layer, the system forms provisional interpretations from those bounded findings. Here the question changes from "what did this sensor output?" to "what local meanings are admissible given the current pattern?" A cue cluster may support a possible drone-like candidate, a possible machinery candidate, an ambiguous moving clutter candidate, or a still-underdetermined anomaly class. Importantly, this layer should preserve ambiguity rather than suppress it. The system does not become more trustworthy by pretending uncertainty has vanished. It becomes more trustworthy by representing candidate meanings in a way that makes later correction possible.

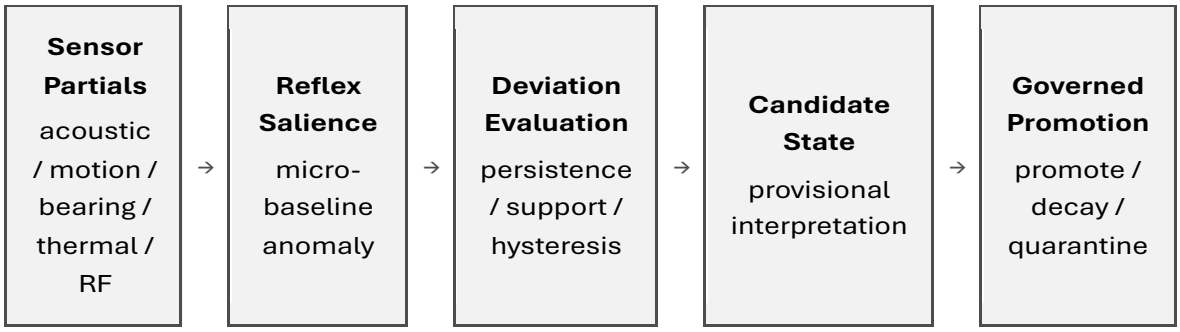
A third layer shapes the immediate response without committing the system to strong belief. This is the response-mediation layer. Its job is not to decide the global truth of the situation. Its job is to shape what happens next within bounded, reversible limits. That may include widening local attention, cueing another sensor, slowing commitment, increasing short-horizon sampling, or preserving the current cue cluster for additional scrutiny. In a clutter-heavy environment, this middle layer is critical because it gives the system a useful way to respond to weak significance without either freezing or overreacting.

The final layer is governance. Governance decides what is allowed to persist, what is allowed to escalate, and what must remain provisional. At this stage, the system may permit promotion from interpreted anomaly to candidate track, hold a case in ambiguous status, quarantine a noisy signal cluster, require multi-sensor confirmation, or deny stronger downstream consequence altogether. The point of governance is promotion control and rationale preservation: provisional signal, no matter how interesting, should not silently become durable state or stronger action without satisfying declared criteria.

A simple pipeline captures the structure: Sensor Partials lead to Reflex Salience; Reflex Salience feeds Deviation Evaluation; Deviation Evaluation supports Candidate State formation; Candidate State is then subject to Governed Promotion. Each step narrows what may be claimed while widening what may be inspected. This prevents the architecture from collapsing into a single opaque score. It also makes the system easier to inspect and improve because local errors can be located at the level where they occur rather than being hidden inside an end-to-end confidence number.

The purpose of stratification is therefore practical rather than philosophical. It gives the system somewhere to hold weak significance before stronger commitment. That is exactly what clutter-heavy, low-altitude sensing requires. Early cues are often too weak to deserve certainty yet too structured to deserve indifference. A layered architecture allows them to matter at the right level and no further.

Figure 1. Stratified weak-signal sensing pipeline for low-altitude small-drone detection



Weak cues influence attention first; stronger commitment requires explicit promotion criteria.

4.1 Minimal Architecture Contract

To keep the proposal concrete, this section defines a minimal contract for each layer in the sensing stack. The purpose of the contract is not to prescribe a single implementation. It is to specify what each layer receives, what it emits, over what horizon it operates, and what it is forbidden to do. This is the minimum needed to make the architecture inspectable and concrete rather than purely rhetorical.

Layer 1: Bounded Observation

Inputs: raw or preprocessed sensor-local features such as acoustic periodicity indices, bearing estimates, motion persistence scores, thermal inconsistencies, or RF-related observables where available.

Outputs: bounded findings only, each with timestamp, source channel, uncertainty estimate, and local feature values.

Time horizon: instantaneous to short rolling window, depending on sensor update rate.

Allowed: preserve local structure, emit uncertainty-bearing findings, remain narrow.

Forbidden: resolved object claims, durable track creation, baseline mutation, or stronger downstream consequence.

Layer 2: Provisional Interpretation

Inputs: bounded findings from one or more channels together with scoped context such as local baseline state, location bucket, or recent horizon summary.

Outputs: hypothesis objects or candidate meanings such as drone-like anomaly, machinery-like anomaly, ambiguous moving clutter, or unresolved weak partial, each with score, supporting cues, and decay conditions.

Time horizon: short rolling horizon across multiple observations.

Allowed: represent ambiguity, score competing candidates, preserve uncertainty.

Forbidden: durable track promotion, policy consequence, or irreversible commitment.

Layer 3: Reflex Salience / Response Shaping

Inputs: hypothesis objects, micro-baseline comparisons, and short-horizon deviation scores.

Outputs: transient salience flags, sensor-cue requests, narrowed inspection windows, temporary weighting changes, or requests for additional evidence.

Time horizon: micro-baseline scale only; explicitly short-lived.

Allowed: raise local attention, adjust next-step sensing behavior, preserve reversible state.

Forbidden: durable memory writes, baseline updates, final object resolution, or stronger downstream authority.

Layer 4: Deviation Governance / Promotion Control

Inputs: deviation metrics over time, cross-channel support, persistence measures, rationale-bearing evidence, and scoped confidence regime.

Outputs: remain provisional, promote to candidate track, decay, or quarantine for audit and later inspection.

Time horizon: longer than reflex, sufficient to evaluate persistence and hysteresis.

Allowed: apply explicit promotion predicates, enforce non-promotion, preserve rationale.

Forbidden: engagement logic, interception decisions, or any operational action outside the sensing architecture.

A minimal hypothesis object in a toy implementation could take the following form:

```
{ id, type, score, supporting_partials[], source_channels[],
first_seen, last_seen, decay_rule, status }
```

A minimal state ladder could then be represented as:

observed → salient → interpreted candidate → governed candidate → decayed / quarantined

The value of this contract is that it makes transitions visible. A reviewer or implementer can now ask concrete questions: What entered the layer? What came out? How long did it persist? What criteria allowed promotion? That is the level at which the architecture becomes testable.

5. Reflex Salience and Micro-Baseline Handling

Many early drone-like cues do not first appear as resolved objects. They appear as short-horizon departures from a recent local corridor. For that reason, the first useful system response is often not classification, but salience elevation. A low-flying drone may present initially as an acoustic irregularity, a directional modulation, a transient but coherent motion trace, or another bounded anomaly relative to what the environment had looked or sounded like a moment earlier. The correct architectural move at that stage is not durable commitment. It is temporary, inspectable attention.

This is the role of reflex salience. Reflex operates over a micro-baseline: a bounded rolling reference for the immediate local stream. The purpose of the micro-baseline is not to establish truth. It is to capture short-horizon expectation well enough that meaningful departures can be noticed quickly. In acoustic sensing, for example, wind, traffic, rotating machinery, insects, and terrain echo may define a local corridor that differs sharply by location, time, and weather. A drone-like cue may therefore be informative not because it matches a universal signature perfectly, but because it departs from the immediate ambient pattern in a structured way.

That localism matters. A weak cue that is highly informative in one environment may be ordinary in another. A periodic acoustic rise near an open rural field may matter differently than the same rise near a construction site. A low-confidence motion trace may be more suspicious in a relatively empty corridor than in a treeline full of small intermittent motion. By comparing new input against a micro-baseline rather than against a static global assumption, the system gains sensitivity without pretending that every anomaly is equally meaningful everywhere.

Reflex salience should also remain explicitly non-persistent. Its job is to modulate near-term attention, not to write durable state. In the present context, that means reflex may raise inspection priority, cue another sensing pathway, tighten a short-horizon sampling window, or preserve a cue cluster for immediate follow-up. What it may not do is silently convert a salience spike into a durable track, a baseline update, or a stronger system claim. That separation is what keeps the architecture from overreacting to clutter.

This is especially important for acoustic and other weak-partial domains. Such cues are often useful precisely because they arrive early, but their early arrival also means they are underdetermined. The architecture should therefore treat a drone-like acoustic irregularity as a bounded salience event first. It matters. It may justify looking harder. It does not yet deserve certainty. In this sense, reflex is best understood as a disciplined middle ground between indifference and premature belief.

6. Deviation Governance Over Multi-Sensor Partials

Once weak partials have been noticed, the next question is not simply whether any one channel has crossed a threshold. The more relevant question is how the emerging pattern departs from the local corridor across time, modality, and scope. This is where deviation governance becomes useful. Rather than treating each cue as an isolated score, the architecture treats the current state as a structured deviation relative to bounded baselines.

In practice, this means that acoustic periodicity, bearing stability, motion persistence, thermal inconsistency, RF-related irregularity, or other available signals can be represented as a deviation structure rather than forced into a single immediate label. A given cue may be weak in one channel and absent in another. That alone does not settle the case. What matters is whether a meaningful deviation vector begins to form: whether the current pattern differs from the recent and scoped expectation in ways that are persistent, convergent, or difficult to explain as ordinary clutter.

This shift in framing matters because many false positives and false negatives occur at the boundary between weak signal and premature commitment. A single anomaly spike may be too weak to trust. A weak but persistent and cross-supported deviation may deserve a different treatment entirely. The architecture should therefore evaluate not only magnitude, but also persistence, cross-channel convergence, and hysteresis. In other words, the system should ask whether the deviation lasts, whether it aligns across channels or nodes, and whether it remains informative after brief fluctuations are damped.

A governed deviation event is not merely a threshold crossing. It is an event formed under declared conditions. Those conditions may include duration, repeated reappearance, directional consistency, cross-sensor support, confidence regime, or other bounded criteria. The exact thresholds are implementation-specific, but the architectural point is stable: the system should not leap directly from a weak cue to a durable claim without a stage that decides whether the pattern has become structurally meaningful enough to deserve broader attention.

Deviation governance should also preserve rationale. If the system elevates a pattern into a candidate event, it should be able to say, in bounded technical terms, what changed: which features departed from baseline, over what horizon, with what degree of confidence, and under which scope. This is not only useful for debugging. It is essential if the architecture is to remain inspectable rather than collapsing into an opaque confidence number that cannot be challenged or improved.

Seen this way, deviation governance is the layer that converts weak but structured change into disciplined attention escalation. It does not yet force a durable track. It does, however, establish when provisional significance has become strong enough to justify more expensive or more coordinated interpretation. That is the point at which the system stops merely noticing and begins formally evaluating whether a track candidate should exist at all.

7. Controlled Promotion: From Cue to Candidate Track

A provisional drone-like cue is not yet a durable track, and a durable track is not yet permission for stronger consequence. This distinction is easy to state and often lost in practice. Once a system becomes interested in a pattern, it is tempting to let interest harden too quickly into commitment. A disciplined architecture resists that temptation by making promotion explicit.

The proposed sensing ladder is simple. A weak partial becomes a salient anomaly. A salient anomaly may become an interpreted candidate. An interpreted candidate may become a governed candidate track. Only after that should stronger downstream routing become eligible, and even then only under declared conditions. The point of this ladder is containment. Each step allows the system to preserve useful structure without letting local attention silently mutate into stronger belief.

Controlled promotion requires more than noticing something twice. Repetition alone should not be sufficient, because repetitive clutter exists. Nor should single-channel salience be enough, because early weak cues are often noisy by nature. Promotion should instead depend on bounded combinations such as persistence across time, convergence across dimensions, directional consistency, or scoped corroboration from another sensing path or node. The exact predicate set may vary by implementation, but the architectural rule remains stable: stronger commitment should require stronger support.

This also means that non-promotion is a first-class outcome. A system should be able to hold a candidate in ambiguous status, decay it safely, or quarantine it as unresolved without treating that as failure. In clutter-heavy environments, good behavior is often not immediate resolution but disciplined refusal to overcommit. A cue that mattered enough to inspect may still fail to earn promotion. The architecture should be able to represent that outcome cleanly.

Promotion, when it does occur, should remain evidence-bearing and inspectable. The system should be able to indicate that a candidate track exists because a bounded set of criteria was satisfied, not because a hidden internal confidence happened to rise. This matters for later debugging, trust calibration, and system comparison. It also matters for safety, because higher downstream consequence should not be resting on an opaque shift in hidden belief.

The practical value of controlled promotion is straightforward. It gives the sensing stack somewhere to hold provisional significance while evidence accumulates, weakens, or dissolves. Without that structure, the system is forced into a bad choice between early certainty and wasted signal. With it, weak cues can remain meaningful without being allowed to become more than they have earned.

8. Why the Sensing Side Resembles Weak-Track Hunting More Than Instant Target Lock

On the sensing side, low-altitude drone detection often resembles weak-track hunting more than instant target lock. The comparison here is structural, not doctrinal. The system is frequently working from distributed partials, cluttered baselines, and gradually accumulating confidence rather than from immediate, high-certainty object identity. That matters because it changes what good early behavior should look like.

In an instant target-lock framing, the ideal is early resolution: see the object clearly, classify it quickly, and move directly toward commitment. In clutter-heavy low-altitude sensing, that expectation is often unrealistic. The earliest cues may be a patterned acoustic disturbance, a bearing that remains coherent over time, a motion trace that persists just long enough to become interesting, or a weak cross-modal alignment that is suggestive without being decisive. The system is therefore often dealing with track quality, persistence, and bounded ambiguity before it is dealing with resolved identity.

That is why a weak-track framing is more useful. It encourages the system to accumulate confidence through structured continuity rather than to force certainty at the first sign of anomaly. It reframes the problem from immediate identification to evidence accumulation: is there enough stable and convergent structure to justify treating the pattern as an emerging candidate rather than as transient clutter? That is a different question, and it leads to a different architecture.

The advantage of this framing is not only technical but behavioral. It makes room for disciplined uncertainty. A good system does not deny ambiguity when ambiguity is structurally present. It manages ambiguity in a way that remains sensitive to weak significance without collapsing into premature commitment. In that sense, weak-track handling is not a downgrade from strong recognition. It is the correct early posture for environments in which early certainty is often unavailable.

This point also helps explain why the paper emphasizes reflex salience, deviation governance, and controlled promotion rather than immediate object claims. These layers are most useful precisely when the system is still operating in a weak-track regime. Their job is to manage the transition from bounded anomaly to candidate track without forcing a false choice between indifference and overconfidence.

9. Toy Simulation Specification

To justify the paper's architectural claims without overstating empirical results, a minimal toy simulation is sufficient. The purpose of the simulation is not to model the full complexity of real-world sensing. It is to create a controlled environment in which weak cues emerge, strengthen, decay, or align across channels so that the proposed layer boundaries can be evaluated.

A minimal simulation can be built around three sensing streams: acoustic, motion, and bearing. The acoustic stream may emit a periodicity index and signal-to-noise delta relative to a local ambient baseline. The motion stream may emit a persistence score and low-confidence track continuity estimate. The bearing stream may emit directional consistency and bearing variance over short windows. Each stream should include noise and ordinary clutter regimes so that weak anomalies must compete against realistic ambient variation rather than appearing in isolation.

A simple timeline is enough. In Phase 1, the environment contains only ambient clutter such as wind-like acoustic variation and intermittent motion noise. In Phase 2, a weak drone-like signature appears acoustically but remains too weak for promotion. In Phase 3, the anomaly persists and bearing coherence strengthens, allowing the architecture to test its transition from salient anomaly to interpreted candidate. In Phase 4, either cross-channel support appears and promotion becomes eligible, or support fails and the anomaly decays. A separate false-positive run should inject a clutter pattern that resembles drone-like acoustics briefly but fails to sustain bearing or motion consistency.

The simulation should evaluate a small set of explicit metrics. At minimum:

- **persistence:** number of consecutive windows above local salience threshold
- **cross-channel support:** number of channels providing nontrivial support within the same horizon
- **bearing variance:** stability of directional estimate over time
- **promotion outcome:** promoted, decayed, or quarantined
- **false-promotion rate in clutter scenarios**
- **missed-promotion rate in sustained weak-track scenarios**

A simple illustrative promotion predicate for the toy environment could be:

Promotion rule:

- persistence ≥ 3 windows
- supporting channels ≥ 2
- bearing variance $\leq$ configured threshold T_b

Otherwise, the candidate remains provisional, decays, or is quarantined depending on rationale and confidence.

The point of including this toy specification is not to claim field readiness. It is to show that the architecture can be instantiated in a narrow, reproducible simulation where reviewers can inspect how weak cues move through the stack. That is enough to support the paper's central claim that early low-altitude drone sensing can be handled as a structured weak-signal interpretation problem rather than as an immediate binary object declaration.

10. Minimal Simulation Use Cases

A useful way to evaluate the proposed architecture is through minimal simulation scenarios in which weak cues emerge, strengthen, or decay under controlled conditions. The point of these examples is not to prove empirical superiority. It is to make the architectural claims concrete enough to test. If the architecture is sound, it should behave differently from both a flat binary classifier and a system that ignores weak cues until certainty is already late.

The first minimal case is an acoustic-only anomaly. An environment contains ordinary ambient variation such as wind and road noise. Within that stream, a periodic deviation appears that differs from the local corridor but remains too weak to justify a durable track. Under the proposed architecture, reflex salience rises, local attention narrows, and the anomaly is preserved for short-horizon follow-up. What does not happen is equally important: the system does not immediately promote the cue into a durable track claim merely because it was interesting once.

The second case adds bearing persistence. The acoustic anomaly reappears over successive intervals and maintains directional coherence. That changes its status. The cue is no longer merely unusual; it is beginning to exhibit structured continuity. At this point the interpretation layer can strengthen the candidate meaning, and deviation governance can evaluate whether persistence and coherence now justify promotion into a candidate track. The architectural behavior to observe here is not sudden certainty, but an orderly transition from weak partial to governed candidate.

The third case is a false positive scenario. An acoustic anomaly briefly resembles a drone-like pattern, but visual, thermal, or motion support fails to appear, and the cue does not remain stable over time. In this case, a disciplined architecture should allow salience to decay rather than forcing commitment. The event should remain inspectable and may still be logged as an ambiguous anomaly, but it should not harden into durable state simply because it briefly crossed a local threshold. This case is important because it shows whether the system can refuse overcommitment cleanly.

The fourth case introduces distributed sensor nodes. Multiple nodes detect low-confidence partials that are individually weak but partially aligned. No single node deserves strong commitment on its own, yet the shared pattern may become more informative when evaluated centrally for temporal and directional consistency. This is where centralized fusion and governed promotion become especially relevant. The test is whether the system can combine provisional cues across nodes without collapsing into either naïve averaging or runaway escalation.

These minimal cases are sufficient for a first simulation-oriented note because they exercise the architecture at its most important boundary: the space between weak significance and strong commitment. A successful run is not one in which every ambiguity disappears immediately. It is one in which weak cues are handled coherently, stronger candidates are earned rather than assumed, and false positives can decay without architectural confusion.

11. Limits and Non-Claims

This proposal is intentionally narrow. It is not a weapons paper. It is not an interception or tactics paper. It is not a claim of deployment readiness. It is not a claim of superiority over current military or commercial systems. It is not a solution to all small-drone detection problems.

Its scope is architectural and defensive. The paper proposes a way to structure weak-signal sensing under uncertainty so that provisional cues can matter without silently becoming durable belief, stronger authority, or high-impact consequence. It is concerned with representation, local interpretation, attentional escalation, deviation governance, and controlled promotion. It does not prescribe engagement policy, hard-kill methods, or operational doctrine.

The paper also does not claim that the proposed architecture eliminates ambiguity. In many low-altitude sensing environments, ambiguity is an irreducible part of the problem. The architectural goal is therefore not total certainty. It is disciplined handling of uncertainty: a way to preserve useful weak structure without overcommitting to it too early.

The proposal is best understood as simulation-oriented rather than field-validated. It is suitable for controlled sensing demos, synthetic sensor fusion studies, and architecture comparisons in environments where clutter, weak partials, and delayed confirmation matter. Any stronger empirical or operational claims would require separate validation, benchmarking, and deployment-specific testing well beyond the scope of this paper.

These limits are a strength rather than a weakness. By keeping the contribution narrow, the paper can make a sharper claim: not that low-altitude drone detection becomes easy, but that weak cues can be handled more coherently when salience, interpretation, persistence, and consequence are explicitly separated.

12. Conclusion

A defensive sensing architecture does not need to be omniscient to be useful. It needs a bounded place for uncertainty while evidence remains incomplete. Weak drone-like cues can matter without immediately becoming truth, track, memory, or action. By separating bounded observation, short-horizon salience, structural deviation evaluation, and governed promotion, the proposed architecture aims to reduce both false indifference and false certainty under clutter-heavy sensing conditions. The contribution is modest but practical: not a claim that small-drone detection becomes easy, but a claim that it can be handled with the same disciplined weak-signal architecture used in other domains where mixed partials, local baselines, and earned scope expansion matter.